Life Sciences and
Facility Management
Institute of
Computational Life Sciences

Image Quality Attributes

Image and Signal Processing

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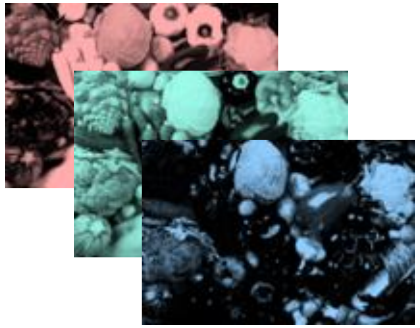
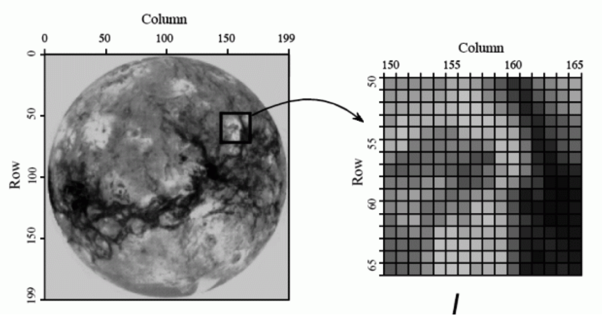
Recap of last week’s topics

- Image processing / computer vision:
  Build machines that can see.

- Image representation
  - Some design choices are inspired by nature (e.g., color representation, bit depth)
  - Images can be represented as matrices

=





Matrix representation

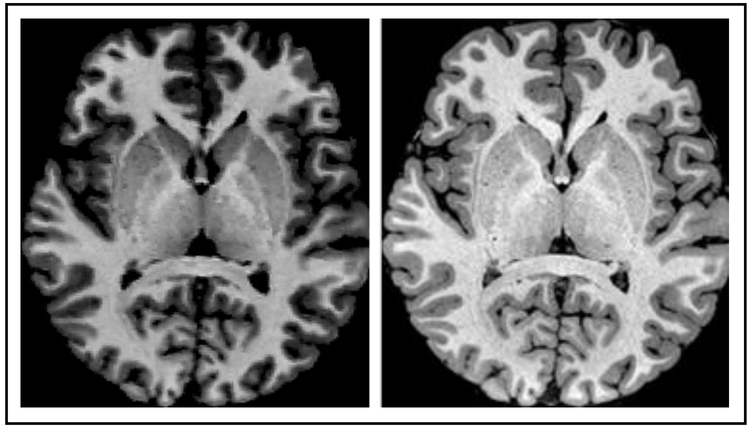
$$\mathbf{B} = (f_{ij}) = \begin{pmatrix} f_{11} & f_{12} & \cdots & f_{1N} \\ f_{21} & f_{22} & \cdots & f_{2N} \\ \vdots & \vdots & & \\ f_{M1} & f_{M2} & \cdots & f_{MN} \end{pmatrix}$$

Illustration: Link

Image quality

Image quality attributes

Task: Examine the image pairs and identify the image quality attributes that influence their appearance.



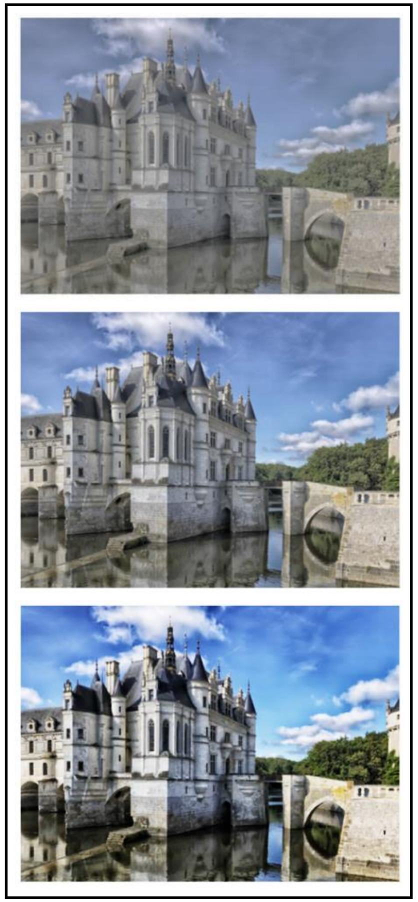
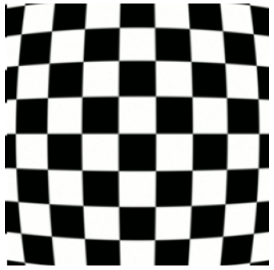
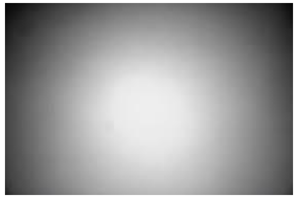


Image quality attributes

- **Sharpness / resolution** determines the amount of detail in an image
- **Pixel noise** refers to random variation of pixel values, obscuring actual information
- **Dynamic range** is the ratio between the brightest and darkest parts of an image
- **Tone reproduction** refers to reproducing correctly the brightness info in an image
- **Color accuracy** refers to the correct representation of colors in an image
- **Contrast** refers to the difference in brightness between different parts of the image
- **Optical distortion** is an aberration that causes straight lines to curve.
- **Vignetting**, refers to the darkening of images near the corners
- **Exposure** refers to the amount of light (per unit area) reaching the photosensor
- **Chromatic aberration** of the lens are color-dependent distortions
- **Aliasing** refers to patterns caused by unfavorable sampling
- **Quantization** may cause visible artifacts like color bands in an image (see exercise)
- ...


Barrel distortion

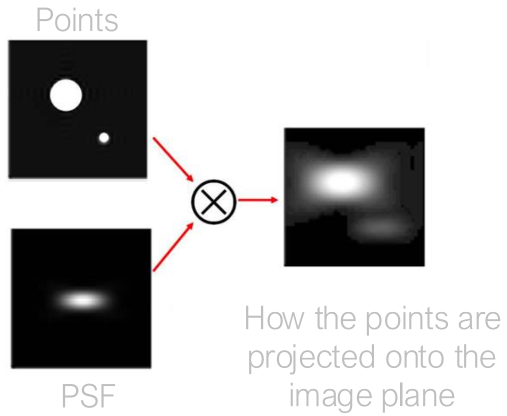

Vignetting

Underexposure

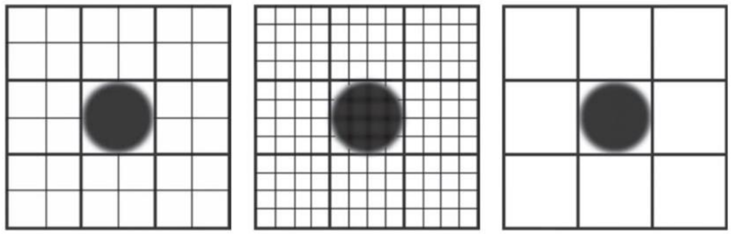

Chromatic aberration

Spatial resolution

- Image resolution refers to the level of detail or clarity in an image. It can be characterized by:
  - Pixel count (number of pixels in the image)
  - Spatial resolution (size smallest discernible detail)
- Spatial resolution is linked to the so-called **point spread function (PSF)** of the imaging system:
  - Describes how a point source appears in the image
  - Equivalent to the impulse response of the imaging system
  - Defines a practical limit for the sensor's pixel size



The imaging system does not capture perfect points. Instead, each point is spread according to the PSF, meaning the image is already blurred before sampling. Therefore, using pixels much smaller than half the PSF width provides no benefit – finer pixels cannot recover information that has already been blurred.



Spatial resolution

- Points with a radius of at least 2 pixels can be reliably reconstructed (even under unfavorable alignment with the pixel grid).
- Recall: The Shannon–Nyquist sampling theorem also applies to spatial sampling.

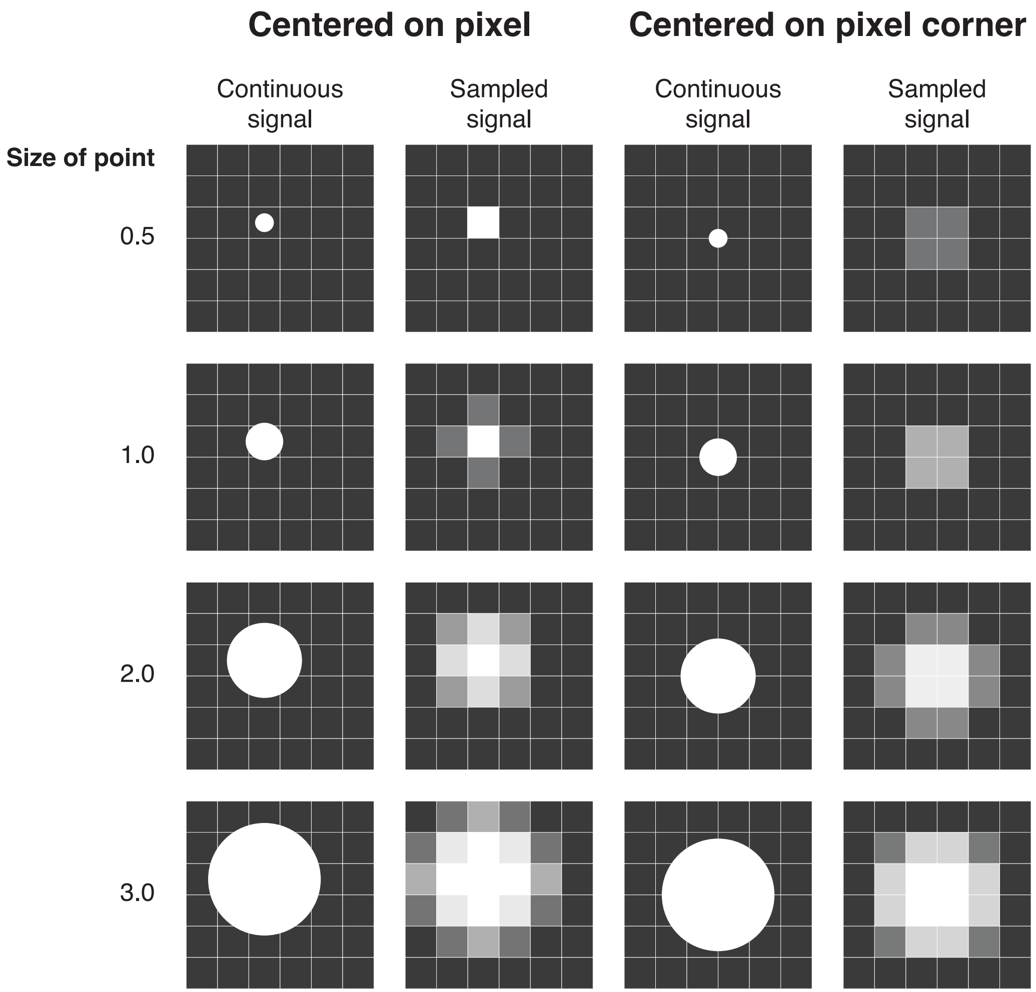


Illustration: Link

Sampling theorem (2D) and aliasing

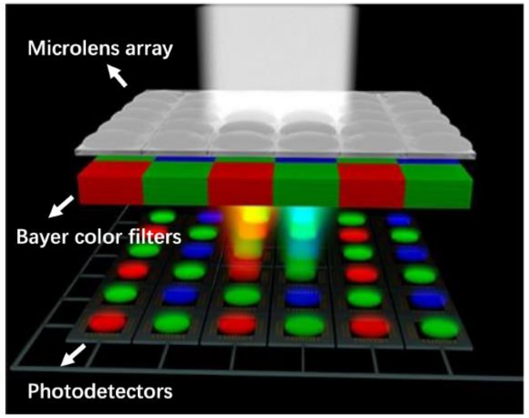
- **Theorem**: If a function $f(x, y)$ contains no frequencies higher than B Hz, then it can be completely determined from its ordinates at a sequence of points spaced less than $B/2$ distance units apart.
- Undersampling leads to **aliasing**, often visible as moiré patterns.
- Aliasing can be reduced by attenuating high-frequency components (low-pass filtering) before sampling.

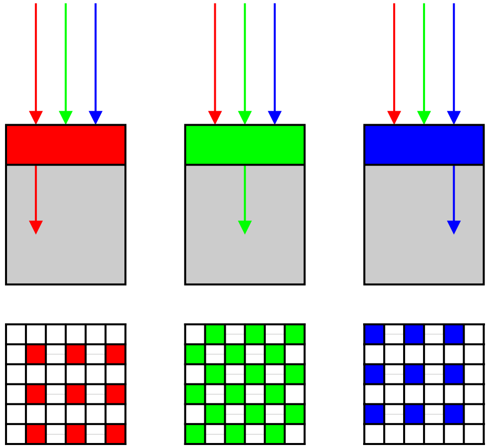

Brick-wall, with and without undersampling. Note: Note: Depending on your display, the right image may also show aliasing artifacts — why?

Illustration: Wikimedia

Color sampling

- Photosensors capture light signals and transform it into digital signals
- Color information is obtained using filters placed over the photodetectors.
- The **Bayer filter** is a common arrangement of these filters (other patterns exist).
- Note: Different colors can be sampled at different rates, which can lead to **color moiré** patterns.



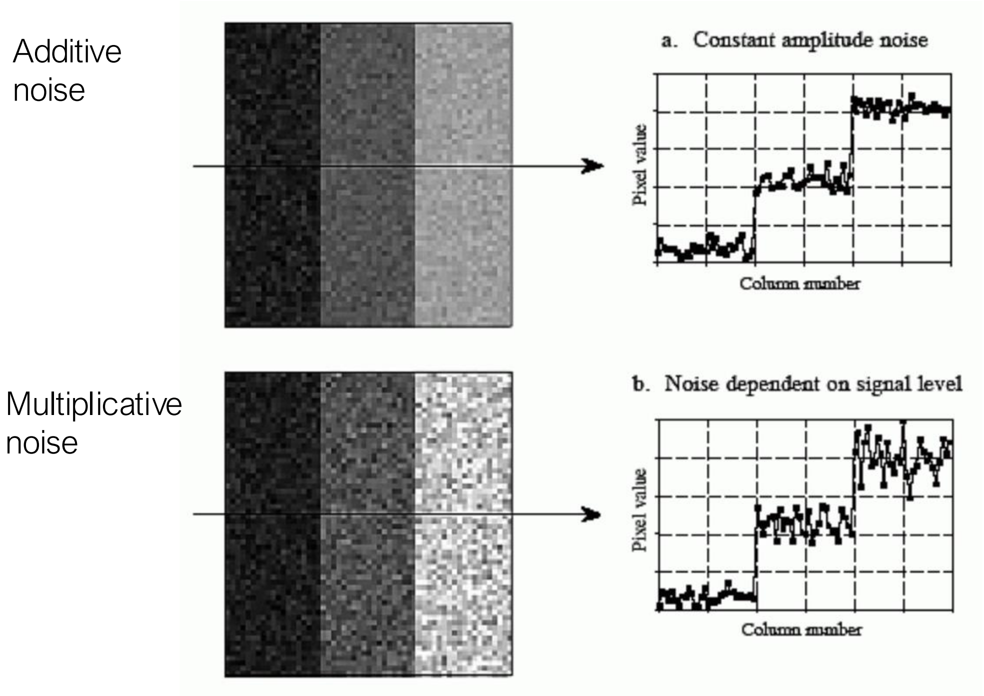


Color moiré pattern

Image noise

- Noise reduces signal quality and degrades image information.

- Types (and causes) of noise:
  - Additive vs. multiplicative noise
  - Gaussian noise (standard model: additive, location- and intensity-independent)
  - Salt-and-pepper noise (with sharp speckles)
  - Frequency-dependent noise / ”colored” noise (white, pink, brown, etc.)
  - Periodic noise
  - …
- Noise affects the **detectability** of information



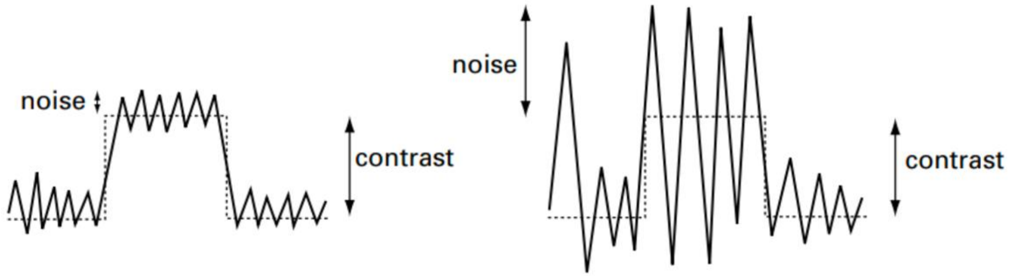


Image noise

Random variable p:
image intensity levels of
a (uniform!) patch

- Signal-to-noise ratio (SNR)
  - Quantifies the level of noise in an image (definitions may vary depending on context)
  - Often evaluated on uniform image patches

$$SNR = \frac{\mu_p}{\sigma_p} \approx \frac{\text{signal amplitude}}{\text{noise amplitude}}$$

- Contrast-to-noise ratio (CNR)
  - Measures the visibility of structures relative to noise → indicator of image quality

$$CNR = \frac{p_{\max} - p_{\min}}{\sigma_p} = \frac{\text{dynamic range}}{\text{noise amplitude}}$$